

A conditional coefficient-prime repair of the five-point height argument

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Abstract

We reconstruct the non-CM arithmetic part of the five-point height argument in the pinned source [1], replacing its new unrestricted 2-adic modularity input by Pan’s published theorem at 5. The 2-adic member of the resulting system supplies the conductor at 5. The deduction is conditional on the preceding geometric and height-comparison inputs stated below; it neither certifies the entire upstream manuscript nor establishes undecidability of rational polynomial solvability.

1 Scope and geometric inputs

Write h for absolute logarithmic Weil height and h_F for the original stable Faltings height. Let $\mathcal{C}$ be the fixed five branch values of the discriminant-210 quaternionic quotient constructed in [1], after its fixed rational coordinate change putting one at ∞ . The distinct involution labels 30, 42, 70, 105, 210, each with a unique branch value, imply individual rationality because the involutions are $\mathbb{Q}$ -defined (lines 162–173). This is that fixed configuration, not a construction for arbitrary five rational points. Choose primitive integral forms $L_b(U, V)$ vanishing at b , with $L_\infty = V$. For $s = (u : v) \notin \mathcal{C}$ put

$$n_{q,b}(s) = v_q(L_b(u, v)) - \min\{v_q(u), v_q(v)\}, \quad M = M_S(s) = \prod_{\substack{q \notin S \\ \exists b: n_{q,b}(s) \text{ odd}}} q.$$

These nonnegative contact orders are invariant under rational rescaling. The product is finite and $M \geq 1$. Enlarge the fixed finite set S to contain 2, 3, 5, 7 if necessary; this changes a claimed height bound only by a fixed constant. Put $T = S \cup \{q : q \mid M\}$.

The following are the reconstructed inputs from [1], `04-height.tex`; their construction is outside the new substitution. All constants below depend only on the fixed data, never on s .

1. A fixed proper cover $\pi : Y \rightarrow \mathbf{P}^1$, with branch index two over $\mathcal{C}$, carries a principally polarized abelian surface family $\mathcal{A}$. For y over s , $A = \mathcal{A}_y$ is defined over a Galois field $L/\mathbb{Q}$ of bounded degree, unramified outside T , and has good reduction over L outside S . All geometric homomorphisms between its conjugates are L -defined; these conjugates are isogenous (lines 303–353 and 573–615).
2. In the non-CM case $\text{End}_{\mathbb{Q}}^0(A) = B$, the fixed indefinite quaternion division algebra of discriminant 210. Thus A is absolutely simple. In the remaining case the preceding CM argument supplies a uniform upper bound for $h_F(A)$ (lines 623–687).
3. The fixed family satisfies

$$h(s) \leq C(1 + \max\{0, h_F(A)\}) \tag{1}$$

(lines 364–371, with its theta-height proof through line 567).

The local reason for the first input is $\pi^*t = z^2$ after an unramified extension outside S : even contact gives unramified lifts. Properness of the spread family gives good reduction even where the fiber field ramifies. The CM case and (1) are retained inputs, not consequences of Pan.

Theorem 1. *Assuming the three inputs above, the non-CM arithmetic argument requires no unrestricted modularity theorem at 2. Published modularity at 5 gives constants $H, c > 0$ such that*

$$h(s) \leq HM_S(s)^c \quad (s \in \mathbf{P}^1(\mathbb{Q}) \setminus \mathcal{C}).$$

2 The prime-independent corrected action

Let $\Gamma = \text{Gal}(L/\mathbb{Q})$. Choose L -defined quasi-isogenies $\mu_\gamma : {}^\gamma A \rightarrow A$, $\mu_1 = 1$, with $\mu_\gamma {}^\gamma b = b\mu_\gamma$ for $b \in B$. Transport of the prescribed B -action gives an automorphism of B ; Skolem–Noether corrects it to this B -linearity. Consequently

$$\mu_\gamma {}^\gamma \mu_\delta = c(\gamma, \delta) \mu_{\gamma\delta}, \quad c(\gamma, \delta) \in \mathbb{Q}^\times.$$

A scalar c on a surface has degree c^4 . With positive real algebraic $a_\gamma = (\deg \mu_\gamma)^{-1/4}$ and $\tilde{\mu}_\gamma = a_\gamma \mu_\gamma$ we obtain the normalized cocycle

$$\tilde{\mu}_\gamma {}^\gamma \tilde{\mu}_\delta = \epsilon(\gamma, \delta) \tilde{\mu}_{\gamma\delta}, \quad \epsilon = c/|c| \in \{\pm 1\}.$$

Here Galois transports the Hom factor and acts trivially on algebraic coefficient scalars. These are the source equations `hei:blinear` and `hei:sign-cocycle` (lines 710–750).

The controlled splitting of lines 766–849 gives a finite Dirichlet character χ , of order $m \leq CM$, and a continuous scalar cochain $\beta : G_\mathbb{Q} \rightarrow \mu_{2m}$ satisfying

$$\beta^2 = \chi, \quad \beta(g)\beta(h)/\beta(gh) = \epsilon(g, h). \quad (2)$$

It is trivial on inertia outside T ; χ is tame at odd primes and has bounded unit conductor at each fixed prime. To recall the control mechanism, identify $H^2(G_\mathbb{Q}, \{\pm 1\})$ with $\text{Br}(\mathbb{Q})[2]$. The class is locally zero outside T . The square-root obstruction of a local finite character is its value at -1 . Odd characters modulo the offending odd primes, and the character modulo 4 at 2, match the finite local invariants; Brauer reciprocity matches the real invariant. Their product has order at most $2 \prod_{q \in T, q \neq 2} (q-1) \leq CM$. Correct a continuous square-root cochain by a sign cochain to obtain (2). A single quadratic character cancels its finitely many unwanted odd-prime inertia signs outside T , introducing possible ramification only at $2 \in S$.

Since the inflated ϵ is one whenever either input lies in G_L , (2) shows that $\beta|_{G_L}$ is a conjugacy-invariant character. Define $G_{L'} = \ker(\beta|_{G_L})$; then

$$L'/\mathbb{Q} \text{ is Galois,} \quad [L' : \mathbb{Q}] \leq 2m[L : \mathbb{Q}] \leq CM. \quad (3)$$

For $\ell = 5, 2$ fix embeddings $\iota_\ell : \overline{\mathbb{Q}} \hookrightarrow \overline{\mathbb{Q}}_\ell$, and define on the extended covariant Tate module

$$R_\ell(g) = \iota_\ell(\beta(g)a_{\bar{g}}) \mu_{\bar{g}} \circ g, \quad \bar{g} = g|_L, \quad R_\ell \simeq r_\ell \oplus r_\ell. \quad (4)$$

The two sign discrepancies cancel, so this is a continuous representation over a finite coefficient extension. Its B -commutation and matrix units in $B \otimes \overline{\mathbb{Q}}_\ell = M_2(\overline{\mathbb{Q}}_\ell)$ give the two-dimensional r_ℓ . This remains valid at 5, which ramifies in B : splitting is over algebraically closed *coefficients*. On $G_{L'}$, R_ℓ is precisely $V_\ell(A)$ with extended coefficients.

3 Pan's theorem and its hypotheses

We use the $p = 5$ specialization of Pan's Theorem 1.0.4 [2] (author version, printed p. 5). A continuous irreducible two-dimensional $\mathbb{Q}_p$ representation of $G_{\mathbb{Q}}$, ramified at only finitely many places, odd, potentially semistable at p , and having distinct Hodge–Tate weights, arises from a cuspidal eigenform up to twist. There is no residual-image assumption for $p \geq 5$. The extra exclusion in Pan's statement concerns only $p = 3$, namely local residual semisimplification of the form $\eta \oplus \eta\omega$.

Each hypothesis holds for r_5 : Faltings' theorems [3] give semisimplicity and $\text{End}_{G_{L'}}(V_5(A)) = B \otimes \mathbb{Q}_5$. Using (4), the commutant of $r_5|_{G_{L'}}$ after scalar extension is the scalars; semisimplicity therefore implies absolute irreducibility, also for r_5 . At 5, A acquires semistable reduction over a further finite local extension of L' . Its Tate module is then semistable, as are its constituents. Hence r_5 is potentially semistable. The Hodge–Tate comparison and finite-extension descent of the de Rham property [4] give cyclotomic exponents 0, 1 for r_5 , each once (cyclotomic Hodge–Tate weight -1).

For complex conjugation j , a scalar $r_5(j)$ would make $R_5(j)$ scalar. Betti comparison identifies $\mu_{\bar{j}} \circ j$ with conjugation followed by a holomorphic quasi-isogeny. It exchanges the two nonzero Hodge subspaces, contradicting scalar action. Since $j^2 = 1$, $r_5(j)$ has eigenvalues 1, -1 and is odd. Outside $T \cup \{5\}$, inertia fixes L , good reduction kills its Tate action, and β is trivial; thus ramification is finite.

Apply Pan. In arithmetic-Frobenius normalization a weight- k form has cyclotomic exponents $0, k-1$; a Tate twist by n shifts both by n . The exponents 0, 1 force $k = 2, n = 0$; finite-character twists may be absorbed into the primitive form. If a dual presentation is used, the identity $V_f^{\vee} \simeq V_f \otimes \psi_f^{-1} \chi_{\text{cyc}}^{1-k}$ gives the same normalization. We obtain a primitive weight-two newform f of level N with $r_5 \simeq V_{f,5}$.

4 Compatibility and the conductor at 5

For $q \notin T \cup \{2, 5\}$, $q \nmid N$, choose arithmetic Frobenius g at q . On the geometric good special fiber $\tilde{A}$ put

$$u_q = \text{red}(\mu_{\bar{g}}) \circ \text{Frob}_q^{\text{rel}} \in \text{End}^0(\tilde{A}), \quad \alpha_q = \frac{1}{2} \beta(g) a_{\bar{g}} \text{Tr}(u_q) \in \overline{\mathbb{Q}}.$$

Clearing denominators, an abelian-variety endomorphism has a rational Tate characteristic polynomial independent of the coefficient prime. Thus $\iota_{\ell}(\alpha_q) = \text{Tr}(r_{\ell}(g))$ for $\ell = 5, 2$. Choose the algebraic conjugate of f selected by ι_5 . Injectivity gives $\alpha_q = a_q(f)$ in $\overline{\mathbb{Q}}$. Chebotarev and semisimplicity imply $r_2 \simeq V_{f,2}$ for the *same* conjugate. This reconstructs lines 992–1038 with $(5, 2)$ in place of $(2, 3)$.

Carayol's conductor comparison [5] applies at $q \neq \ell$, including monodromy. Accordingly use r_5 at every $q \neq 5$, and r_2 at $q = 5$. Outside T the conductor exponent is zero. At large q , larger than $[L : \mathbb{Q}]$ and all fixed exceptional primes, wild inertia fixes L and acts trivially on the prime-to- q Tate module. Its β -image is sign-valued because $\beta^2 = \chi$ is tame. A pro-odd group has no nontrivial sign quotient; hence r_{ℓ} is tame and has conductor exponent at most two.

For completeness the bounded small-prime argument also covers 5. For fixed q and $F = L_v$, bounded degree leaves finitely many local fields. The square of $\beta|_{G_F}$ corresponds to $\chi \circ N_{F/\mathbb{Q}_q}$, with uniformly bounded unit conductor. On sufficiently deep units U_F^b , β is sign-valued. Logarithmic coordinates show $(U_F^b)^2 = U_F^b$ for odd q , or $U_F^{b+v_F(2)}$ at 2, so β is uniformly trivial sufficiently deep. Choose fixed $p_0 \geq 3$, $p_0 \neq q$. Over $F_1 = F(A[p_0])$ inertia is unipotent [6]; $[F_1 : F] \leq |\text{GL}_4(\mathbf{F}_{p_0})|$. A compact unipotent ℓ -adic group has no nontrivial pro- q subgroup for $\ell \neq q$. The finite local-field choices and Herbrand rules therefore give a uniform upper ramification cutoff for r_{ℓ} . Its Swan conductor and dimension-two inertia contribution are bounded. The fixed small primes contribute a constant, proving

$$N \leq CM^2. \tag{5}$$

This argument uses bounded $[L : \mathbb{Q}]$, not the growing degree of L' .

5 The modular factor and five-point bound

Let $J = J_1(N)$, $g_0 = \dim J$. Its 5-adic module contains r_5 . Over $G_{L'}$, $V_5(A)$ consists of two copies of r_5 , so an extended-coefficient equivariant Hom to $V_5(J)$ is nonzero. In finite dimension finitely many commutation equations cut out the invariant Hom space; their kernels commute with scalar extension. Faltings thus gives an actual nonzero L' -map $A \rightarrow J$. Absolute simplicity makes its kernel finite; complete reducibility gives

$$J \sim_{L'} A \times D_0.$$

The source's Belyi-degree estimate and Javanpeykar's height bound [7], in the same stable-height convention, give $g_0 \leq CN^2$ and $\max\{0, h_F(J)\} \leq CN^{12}$.

Zarhin's trick [10] gives a principally polarized complement $D_Z = (D_0 \times D_0^\vee)^4$, and $C_1 = J^8 \sim_{L'} C_2 = A^8 \times D_Z$. Set $g = 8g_0$ and $d = [L' : \mathbb{Q}]$. By (3)–(5), $g \leq CM^4$, $d \leq CM$, and $\max\{0, h_F(C_1)\} \leq CM^{24}$. Gaudron–Rémond [8] gives an isogeny $\phi : C_1 \rightarrow C_2$ with

$$\log \deg \phi \leq 1024g^3 \left(64g^2 \log(14g) + \log d + 2 \log \max\{h_F(C_1), \log d, 1\} \right) \leq CM^C.$$

Crucially its source height is already known. Additivity and the isogeny-height inequality imply $8h_F(A) + h_F(D_Z) \leq 8h_F(J) + \frac{1}{2} \log \deg \phi$. Bost's dimension-proportional lower bound [9] gives $h_F(D_Z) \geq -C \dim D_Z$ in this normalization. Consequently $h_F(A) \leq CM^C$. Combine with (1); the retained uniform CM bound covers all other fibers and $M \geq 1$ absorbs constants. This proves Theorem 1.

Status and disclosure. The new substitution is the choice of coefficient primes (5, 2) and the application of published Pan modularity; the geometric and height inputs and scalar descent are reconstructed from the pinned manuscript. This conditional supplement does not use the direct fixed-field von Känel–Kret shortcut and does not settle the separate pointwise 2-converse dependency. AI tools were used in drafting, source inspection, and adversarial checking; these checks are not conventional peer review.

References

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